

Abortion Access and Female Sterilization

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Abstract

In the wake of the *Dobbs v. Jackson* decision, an evolving literature has turned to the question of how individuals might mitigate the negative consequences of restricted access to abortion through changes in their contraceptive decision making. In this paper, we exploit a natural experiment provided by House Bill 2 (HB2) in Texas to measure the effects of abortion access on rates of female sterilization in the state. HB2 passed in July 2013, imposing strict regulations on abortion providers and shuttering over half of all abortion clinics in Texas within months. We combine information on the dramatic increase in county-level travel distance to an abortion provider with inpatient claims data to determine how rates of tubal ligation evolve in Texas following HB2. Additionally, we explore how effects on sterilization are heterogeneous by race, age, and other pregnancy/birth-related health outcomes.

1. Introduction

For individuals avoiding pregnancy, choosing a contraceptive method balances effectiveness, health risks, and desire for future children. Most contraceptive methods involve some risk of pregnancy. 13% of condom users, for example, will become pregnant within the first year of use. In contrast, sterilization procedures, including tubal ligation or female sterilization, reduce the one-year pregnancy risk to just 0.5% (Guttmacher Institute, 2020). However, sterilization is a permanent procedure, and therefore individuals choosing sterilization will not be able to have more children in the future.

The 2022 *Dobbs v. Jackson* ruling resulted in strict, state-level abortion bans, potentially changing the calculus of contraceptive choice. Access to abortion provides a safety net in cases of contraceptive failure. Media reports feature anecdotes of women considering permanent sterilization procedures along with data demonstrating increases in sterilization rates after *Dobbs* due to increased anxiety around unintended pregnancy (Healthline, 2024; Bolton, 2022). Studies show that more women under 30 are choosing permanent contraceptive methods (Ellison et al., 2024) suggesting that, in the face of limited abortion access, some women prefer irreversible fertility loss in exchange for reduced pregnancy risk.

To better understand the relationship between abortion access and changes in demand for tubal sterilization, this paper uses variation in abortion restrictions resulting from policy changes in Texas. In 2013, Texas passed House Bill 2 (HB2), legislation that imposed strict targeted regulations on abortion providers, also known as TRAP laws. The regulations ultimately shuttered more than half of the abortion-providing clinics in the state, limiting access for many individuals throughout Texas. Using administrative inpatient records, we leverage this shift in abortion access to identify changes in demand for tubal sterilization and investigate the mechanisms underlying these changes. We develop a theoretical framework to model the decision to pursue a tubal ligation and the mechanisms behind this decision. Our work provides evidence consistent with more women choosing sterilization following an unintended birth.

We use difference-in-differences and event study analyses to measure the impact of increasing travel distance to an abortion provider on rates of tubal ligation in Texas between 2011 and 2015. We find that 6 months following a 100-mile increase in travel distance, tubal ligation rates increase by 9%. This six-month lag provides evidence of a behavioral shift among women who discover an unintended pregnancy in their first trimester just as the first clinics

closed in response to HB2. This finding is consistent with a scenario where women are forced to carry an unintended pregnancy to term and subsequently undergo a postpartum sterilization at delivery. Heterogeneity analysis reveals that this behavioral shift toward permanent contraception is likely concentrated in the more economically advantaged counties. Placebo analysis indicates that the increase in tubal ligations is not caused by simultaneous increases in pregnancy or delivery complications. Together, our results indicate that women in Texas do receive more tubal sterilizations due to limited access to abortion following the policy change.

The *Dobbs* decision led to an influx of research examining the relationship between abortion restrictions and fertility and contraceptive behavior. Dench et al. (2024) find that abortion bans due to *Dobbs* led to a 2.3% increase in births, with larger concentrations among Hispanic women and women aged 20-34. Related literature identifies a sharp increase in permanent contraceptive methods immediately following *Dobbs* (Ellison et al., 2024). We extend this literature by examining demand for tubal sterilization following abortion restrictions using granular county-level data on abortion access due to changes in travel distance to abortion-providing clinics. Our work provides evidence that increases in female sterilization are consistent with behavioral responses to unintended births carried to term in the absence of abortion access.

Research on the effects of *Dobbs* builds upon prior research examining abortion access, unintended births, and economic equality. Early work demonstrates that fertility control enabled by the birth control pill spurred years of economic advancement among women (Bailey, 2005, 2010, 2013; Goldin & Katz, 2002). A large body of literature further finds that access to legal abortion leads to a decline in fertility, delays in marriage and motherhood, improved educational attainment, and improved health outcomes for women (Farin et al., 2021; Jones & Pineda-Torres, 2024; Levine et al., 1999; Myers, 2017). In contrast, restricted abortion access results in increases in maternal mortality and morbidity (Gardner, 2024; Kheyfets et al., 2023). Our work emphasizes how abortion restrictions reshape contraceptive choices, suggesting that prior estimates of fertility and health may understate the full response to abortion restriction because some individuals shift toward permanent sterilization.

The sharp reduction in abortion access following HB2 generated substantial research on the effects of abortion restrictions in Texas. This paper contributes to that literature by examining changes in demand for permanent sterilization. Prior work identifies increases in birth rates within the state and decreases in abortions (Fischer et al., 2018; Lindo et al., 2020). Researchers

also examine how HB2 changes contraceptive choice. Ahmed et al. (2025) found spikes in Google searches for sterilization procedures in the months post-Dobbs. While Fisher et al. (2018) finds no change in emergency contraceptive and condom sales in response to the legislation, Crowe et al. (2025) finds an increase in intrauterine devices (IUD) of 0.55 IUDs per 10,000 reproductive age females per quarter in counties that lost access to an abortion provider. The authors, however, do not identify a change in vasectomies. A similar paper, using data on active-duty servicepeople in the U.S., found increases in both vasectomy and tubal ligation procedures post-Dobbs, though these results may not be generalizable to the non-military population (Koehlmoos et al., 2025).

In the next section, we explain the relevant history and background of the policy environment and a description of the benefits and risks associated with tubal sterilization. Section 3 describes the data used in this paper and Section 4 presents the methods and results. Section 5 concludes.

2. History and Background

2.1. Historical Context

Texas House Bill 2 (HB2), originally proposed as Senate Bill 5 (SB5), emerged during a tense, anti-abortion political climate in the state. The legislation followed a period of budget cuts targeting women's health initiatives and family planning clinics, which had already reduced access to reproductive health services throughout the state (Fischer et al., 2018; White et al., 2015). During this time, Governor Rick Perry's stated goal was to "make abortion at any stage a thing of the past," and his sentiments were heavily covered by local, national, and state media outlets (CBS News, 2012; Bassett, 2012.; Rayfield, 2012).

Outlawing abortion proved to be more difficult than anticipated and multiple bills failed during the regular legislative session. In May 2013, Perry ordered a special session where SB5 was introduced in the Texas Senate, ultimately failing to pass. A second special session began in July 2013, where SB5 was reintroduced as HB2, passing in both houses and signed into law on July 18.

HB2 included multiple TRAP laws, which required that doctors performing abortions have admitting privileges at local hospitals within 30-miles of the clinic and that individuals taking abortion-inducing medications have medical oversight. The law also required that facilities providing abortions meet the standards of an ambulatory surgical center, such as

minimum door and hallway widths, surgical grade ventilation, sterile operating rooms, backup power systems, and specialized credentialing for staff (Center for Medicare & Medicaid, 2024).

These requirements forced small reproductive clinics to either undergo expensive upgrades, stop providing abortions, or close their doors entirely to comply with regulations. The law went into effect in November 2013, and between 2011 and 2014, 25 abortion facilities closed, leaving only 17 clinics remaining in the state. The closures fundamentally changed abortion access in Texas, mainly impacting residents in rural areas (Crowe et al., 2025). Many residents faced over 100 miles of travel to the nearest clinic, along with congestion and long wait times due to reduced access (Lindo et al., 2020).

2.2 Tubal Sterilization

Tubal sterilization broadly includes surgical procedures designed to permanently prevent pregnancy either by closing or severing the fallopian tubes or removing them entirely (Planned Parenthood, 2026). This paper focuses on the subset of tubal ligation procedures involving severing or closing the fallopian tubes, excluding procedures like bilateral salpingectomy, which is the removal of the fallopian tubes. Tubal ligation procedures include laparoscopic sterilization, laparotomy, and mini-laparotomy. Laparoscopic sterilization can be completed as an outpatient surgery at any time and involves sealing the fallopian tubes with a band, ring, or clip. The laparotomy and mini-laparotomy are more extensive surgeries requiring an inpatient hospital stay. Because they are more invasive procedures, laparotomy and mini-laparotomy are typically completed after abdominal surgery (like a C-section) or vaginal childbirth, respectively (Cleveland Clinic, 2023).

Tubal ligation procedures are some of the most expensive contraceptive methods, with estimated surgical and prescription costs averaging \$7,568 among commercially insured people (Carney et al., 2017). This figure accounts for all claims billed to insurance companies and may not reflect the out-of-pocket costs to patients. As with other forms of contraception, tubal ligation coverage is mandatory at 100% coverage under the Affordable Care Act with no co-pays or deductibles, and the coverage includes all services related to the procedure. Thus, patients with commercial insurance face minimal financial costs for tubal ligation. However, current Medicaid policy in Texas does not reimburse for tubal ligations for women under 21 (Potter et al., 2021a).

In addition to financial costs, patients interested in tubal ligation face medical risks and discomfort related to the procedure. All tubal ligation procedures are considered major surgeries

and involve pain, swelling, and post-operative recovery time, along with risk of complications including infection or injury to other organs (Cleveland Clinic, 2023). The procedures are completed under anesthesia, which also involves risks and side effects including throat pain from breathing tubes, nausea, cramping, bloating, and dizziness. In addition, while some forms of contraception reduce or eliminate menstruation, tubal ligation does not stop menstrual cycles and may be associated with heavier bleeding or menstrual pain.

Because tubal ligation is intended to be permanent, choosing the procedure involves different long-term considerations than reversible contraceptive methods. A study by Shreffler et al. (2016) finds that 72% of women do not regret tubal ligations, though women who undergo sterilization at a younger age are more likely to express regret than older women (Hillis et al., 1999). Though the procedure is meant to be permanent, it can be reversed with some success and Black/Hispanic women are more likely to request a reversal than white women (Borrero et al., 2011). Having fewer children or a change in marital status also are associated with regret (Shreffler et al., 2016).

Rates of tubal ligation also vary by demographic. Black and Hispanic women are more likely than white women to undergo tubal ligation, as are women with no or public insurance (Borrero et al., 2006). Controlling for race, religion, and socioeconomic factors, rural women are about twice as likely to be sterilized than urban and suburban women (Clark and Levy, 2025; Travers et al., 2017), though urban women are most likely to request a reversal. Researchers hypothesize that the higher demand for sterilization in rural areas is a problem of access: rural women are less likely to receive care from reproductive health services or dedicated OBGYNs, and rural physicians generally have less training on the insertion of other LARCs (Clark and Levy, 2025).

Tubal ligation is one of the most effective contraceptive methods, leading some women to choose sterilization for reasons beyond pregnancy prevention. Shreffler et al. (2016) find that nearly half of the women (46%) surveyed in their sample reported motivations unrelated to contraception when electing tubal ligation. Common motivations include medical concerns, such as reproductive health conditions or genetic disorders that individuals do not wish to pass on to future children. Other factors include financial circumstances, age, and pressure from family members, partners, or physicians.

Cost- and systems-level barriers, such as issues with the Medicaid consent form, can prevent women from receiving tubal sterilization. Research suggests a level of unmet demand for tubal sterilization among low-income women in Texas. Potter et al. (2021b) surveyed postpartum women who indicated they did not want more children and wanted a tubal sterilization. The authors found 48% of these women did not obtain a tubal sterilization before they were discharged from the hospital after giving birth. Among the women who were unable to obtain a tubal ligation, some became pregnant, with 12.5% of women carrying to at least 12-weeks gestation within 8 months after their initial pregnancy. 59% of these pregnancies were reported to be unwanted.

Ultimately, medical risks, administrative hurdles, and potential for future regret acts as a high barrier that may deter women from choosing permanent sterilization. In the pre-HB2 landscape in Texas where abortion was accessible, an individual could decide the costs of an intensive medical procedure did not outweigh the risk of contraceptive failure, given the availability of abortion as an insurance mechanism. However, without the option of abortion, the risk of contraceptive failure shifts from seeking an abortion to facing an unintended pregnancy. In the remainder of the paper, we evaluate whether individuals respond to this change by seeking permanent contraception.

3. Theoretical Framework

To understand how abortion restrictions shift an individual's preferences for permanent contraception, we build upon the self-protection and self-insurance model derived by Ehrlich and Becker (1972) where abortion acts as insurance against contraceptive failure. Agents are people with the capacity to become pregnant who do not want children at time $t + 1$. Agents are risk-averse and face disutility associated with an unintended pregnancy, equal to V , which varies based on an individual's access to abortion.

In this setting, there are two regimes: the "accessible regime" representing pre-HB2 and the "restricted regime" representing post-HB2. In the accessible regime, the agent's disutility associated with an unintended pregnancy is equal to V_a , which includes all the costs associated with an abortion. In the restricted regime, the individual faces disutility of V_b , which are the costs associated with an unintended birth. If an agent prefers abortion over carrying a pregnancy to

term, then $V_a < V_b$. Individuals for which $V_a > V_b$ prefer carrying an unintended pregnancy to term over abortion, and therefore their contraceptive choice is unaffected by abortion access.

Agents choose from a menu of contraceptive options at time t , each with varying degrees of effectiveness to protect against the risk of an unintended pregnancy. For simplicity, allow there to be three contraceptive options: $c = \{0, 1, 2\}$, where $c = 0$ is a low-efficacy option, such as no contraception, $c = 1$ is a highly effective, reversible option, such as an IUD, and $c = 2$ is a permanent, irreversible option, such as a tubal ligation. Each contraceptive option is associated with a probability of failure, $p(c)$, where $p(0) > p(1) > p(2)$ and $p(2) = 0$. Ultimately, at time t the individuals risk exposure is:

$$E[V] = p(c) \cdot V$$

In addition to variation in risk, each option is associated with certain costs, K_c , including procedure risks, logistical costs (such as travel distance), and administrative frictions, which vary with each contraceptive option.

For simplicity, we assume K_0 and K_1 are constant and K_2 is state-dependent based on whether the agent is giving birth. As described above, tubal sterilization procedures commonly occur alongside C-section and vaginal deliveries, minimizing the frictions that occur with these procedures. Therefore, for an agent who is not pregnant, $K_{standalone}$ is maximized and includes the costs and risks associated with undergoing a major medical procedure. In contrast, these barriers drop substantially for postpartum agents since hospitals can perform the procedure with the delivery. Therefore, $K_{postpartum} < K_{standalone}$.

This theoretic framework leads to two hypotheses:

Hypothesis 1: Defensive contraception.

Following Ehrlich and Becker (1972), moving from the accessible to the restricted regime removes insurance against unintended pregnancy, driving agents to shift their demand toward more effective contraceptive methods, including sterilization, in anticipation of an unwanted pregnancy. Because abortion is no longer an option, there is an increase in disutility if contraception fails, such that $p(c)V_a < p(c)V_b$. Forward-looking agents will invest the transaction costs, K_c , into more effective contraceptive options if the cost of adoption is less than the expected risk exposure:

$$K_c < \Delta p(c) \cdot V_b$$

Where $\Delta p(c)$ is equal to the reduction in the probability of failure between the contraceptive options. This hypothesis is supported by Crowe et al. (2025), who find an increase in claims for IUD's for the Texas counties that lose access to abortion, aligning with the set of individuals who switch from $c = 0$ to $c = 1$.

Because sterilization drives the risk of pregnancy to zero, $p(2) = 0$, the risk reduction of choosing sterilization over the baseline method c is $p(c)$. Therefore, if the cost of a sterilization procedure, $K_{standalone}$, is sufficiently low, then we expect to find an immediate spike in tubal ligations following HB2.

Hypothesis 2: Responsive sterilization.

The fixed cost of sterilization as a standalone procedure is quite high and involves potential time off work, surgical risks, and potential negative side effects, which may deter agents from choosing standalone sterilization. However, if an agent experiences an unintended pregnancy due to contraceptive failure in the restricted regime, they now must carry the pregnancy to term. When they give birth, the cost of sterilization is $K_{postpartum}$, which is significantly lower than $K_{standalone}$ because they are already hospitalized and under the care of a gynecological team. In this scenario, we expect to find a lagged spike in tubal ligations after HB2, with the majority of procedures occurring after a delivery.

4. Data and Summary Statistics

To identify cases of tubal ligation, we utilize publicly available healthcare claims data from the Texas Health Care Information Collective Inpatient Public Use Data File (THCIC, 2011-2015). THCIC compiles healthcare claims (invoices submitted to insurance companies) reported by Texas hospitals and makes the de-identified data available for public use. The data include rich details on each patient claim, such as the associated diagnosis and procedure codes, the year and quarter the claim was submitted, and patient demographic information including gender, race, and county of residence. All state-licensed hospitals are required to report claims to THCIC, with limited exceptions.¹

¹ Small, rural hospitals such as those located in counties with a population of less than 35,000 or those with fewer than 100 hospital beds are not required to report. Additionally, facilities that do not seek insurance or government reimbursement are not required to report.

We identify 155,262 tubal ligations between 2011 and 2015 through the following International Classification of Diseases, 9th Revision (ICD-9) codes: 66.21/66.31 bilateral ligation and crushing of fallopian tubes, 66.22/66.32 bilateral ligation and division of fallopian tubes, 66.29/66.39 bilateral destruction or occlusion of fallopian tubes. Because inpatient stays often involve multiple procedures, the inpatient data includes 25 different surgical procedure code variables: a primary surgical procedure code and 24 “other” surgical procedure codes. We identify a tubal ligation procedure if the aforementioned ICD-9 codes appear in any of the 25 procedure code variables. Of the 155,262 records, only 15,776 of appear as the primary surgical procedure code. The majority of tubal ligation procedures in our sample happen in the same discharge record as a C-section delivery, as shown in Table 1.

Table 1: Most Common Primary Surgical Procedure Codes for Tubal Ligation Sample

Primary procedure code	Description of procedure code	# of records	% of records
741	low-cervical C-section	111,895	72.07
7359	manually assisted delivery	16,589	10.68
6632	bilateral ligation and division of fallopian tubes	10,384	6.69
6639	bilateral destruction or occlusion of fallopian tubes	4,897	3.15
7569	repair of current obstetric laceration	4,832	3.11

Table 2 provides demographic details about the Texas population who received a tubal ligation between 2011 and 2015. In this population, tubal ligations are most common among women aged 25-39. Hispanic women are slightly overrepresented in the sample; while 40.2% of the Texas population is Hispanic, 47% of the sample is Hispanic. Simultaneously, white women (39% of the population) and Black women (13.4% of the population) are underrepresented in the sample.

Medicaid is most frequently listed as the primary payer for the discharge records that include tubal ligation. One limitation of the payment source component of the THCIC data is that the “primary payer” refers to the payment source contributing the most to the entire set of

procedures in the discharge record. Because many tubal ligations are performed postpartum and Medicaid finances approximately 40% of births in the United States, Medicaid is most commonly the primary payer in these records. Pregnancy-related Medicaid eligibility thresholds are more generous than the standard Medicaid eligibility thresholds, extending coverage to many low-income pregnant women who would not otherwise qualify for Medicaid.

Table 2: Summary Statistics

Variable	# of records	% of records	N
<u>Age</u>			155,262
15-17	11	0.01	
18-19	81	0.05	
20-24	18,132	11.68	
25-29	42,356	27.28	
30-34	49,474	31.86	
35-39	34,905	22.48	
40-44	9,848	6.34	
45-49	455	0.29	
<u>Sex</u>			155,262
Female	155,261	100	
Unknown	1	0	
<u>Race/Ethnicity</u>			153,979
Non-Hispanic White	50,976	33.11	
Non-Hispanic Black	17,250	11.20	
Hispanic	72,365	47.00	
Other	4,629	3.01	
Unknown	8,759	5.69	
<u>First Payment Source</u>			155,203
Medicaid	82,430	53.11	
PPO	17,497	11.27	
HMO	16,989	10.95	

Blue Cross/Blue Shield	12,997	8.37
Commercial Insurance	12,791	8.24
Charity, Indigent, or Unknown	3,845	2.48
Other	8,654	5.58

We determine how HB2 changed access to abortion using a panel describing the distance to the nearest abortion provider from Myers (2025). The Myers data calculate the distance in miles from each county population centroid to the nearest abortion-providing facility in each month from 2009 through 2025, allowing us to observe changes in travel distance to the nearest abortion provider following HB2-related clinic closures. We average these distances across quarter-years and then match the aggregated distance measure to our tubal ligation data for Texas counties between Q1 of 2011 and Q3 of 2015.² To determine the rate of tubal ligations, which is our outcome variable, we use population data from SEER, allowing us to calculate the number of tubal ligations per 10,000 females.

Figure 2 describes the raw correlation between changes in travel distance following HB2 and changes in the county-level rate of tubal ligation. A 100-mile increase in travel distance is associated with an increase in the rate of tubal ligation by 0.29 procedures per 10,000 females. This correlation is statistically significant at the 10% level and represents a 4.7% increase from the average rate of tubal ligation (6.09). Though this represents a purely descriptive relationship, the figure is consistent with the hypothesis that reduced access to abortion may increase demand for permanent sterilization.

To further examine changes in the rate of tubal ligation procedures over time, we compare counties plausibly affected by HB2 to counties where travel distance was unaffected by the policy change. Figure 3 provides population-weighted trends in tubal ligation for counties that experience more than a 30-mile increase in travel distance after HB2 was passed in July 2013 (n=106) compared to counties that experience no change in travel distance following the law (n=59). Leading up to the policy change, trends are nearly identical across the two groups of counties. After HB2 is passed, however, tubal ligation rates increase in counties with large travel distance changes relative to unaffected counties, suggesting a shift in tubal ligation demand

² We exclude 2015Q4 from the analysis because THCIC transitioned from ICD-9 to ICD-10 codes in 2015Q4. While ICD-10 also identifies tubal ligation, the procedure codes are not directly equivalent.

alongside changes in abortion access. The figure also shows seasonal trends in tubal ligation rates that follow seasonality in birth rates, which generally peak during the summer months.

Figure 2: Association Between Changing Travel Distance and Rates of Tubal Ligation

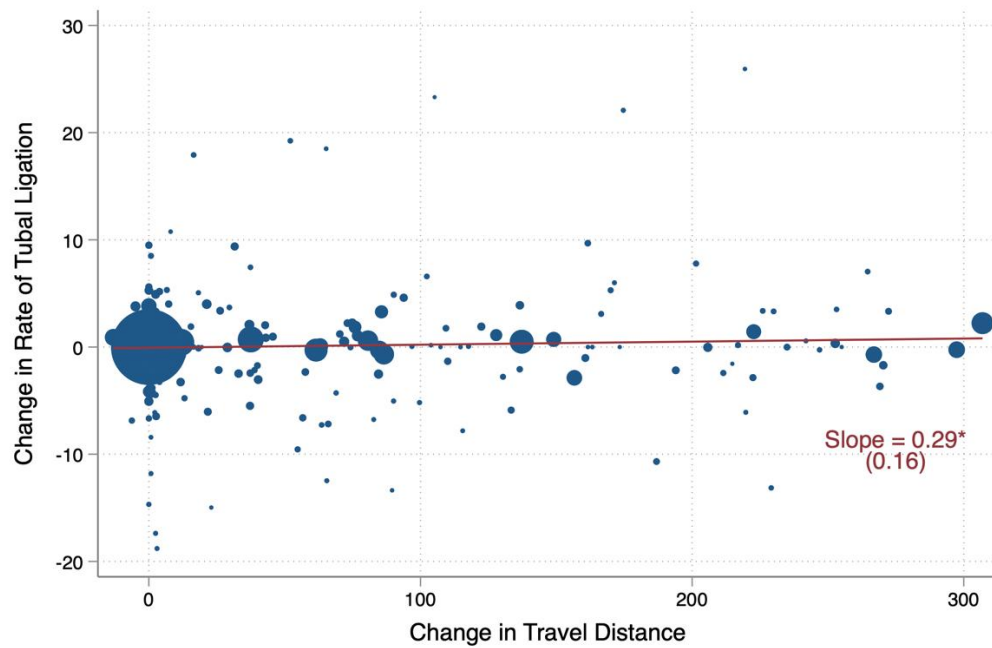
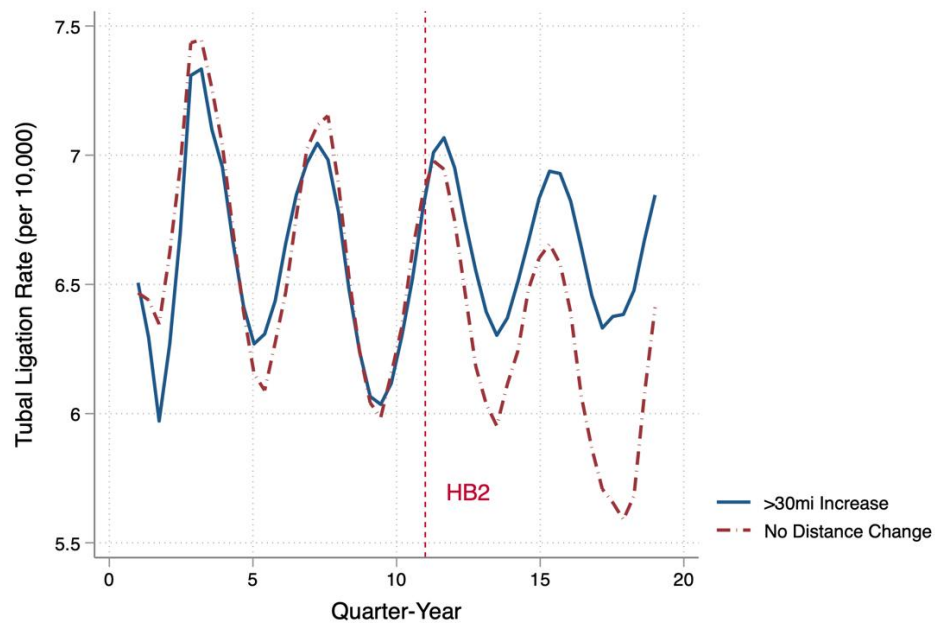


Figure 3: Trends in Rate of Tubal Ligation



5. Analysis

To measure the causal effect of reduced abortion access on rates of tubal ligation, we use a continuous two-way fixed effects (TWFE) difference-in-differences design. Equation (1) describes the estimating equation, where Y_{ct} is the rate of tubal ligation in county c in quarter-year t , α_c and δ_t are county and quarter-year fixed effects respectively, and X_{ct} represents a set of time varying county-level economic control variables including: median household income, unemployment rate, per capita personal income, and the poverty rate. The travel distance treatment variable, td_{100} , represents travel distance in 100's of miles. The specification includes a linear and quadratic term for the continuous treatment measure to capture any nonlinear effects of larger travel distance increases.

$$Y_{ct} = \alpha_c + \delta_t + \beta_1 td_{100} + \beta_2 td_{100}^2 + \beta_3 X_{ct} + \epsilon_{ct} \quad (1)$$

β_1 captures the causal effect of an additional 100 miles of travel to an abortion provider on the rate of tubal ligation, conditional on the assumption that rates of tubal ligation would have trended similarly in treated counties absent the travel distance change. In all specifications, observations are weighted by the population of all females in the county and standard errors are clustered at the county level.

To test if increases in tubal ligation following HB2 operate through the responsive sterilization channel, or in response to a contraceptive failure, Table 3 provides a series of estimations of Equation (1) including a set of lags and leads of the treatment variables. Future changes in travel distance should not affect tubal ligation in the current period, supported by the estimates presented in Columns (1) and (2).

However, if abortion policy affects tubal ligation rates through the channel of increased contraceptive failure and therefore increased births, we expect a delay in observing the increases in tubal ligations. The lags of the treatment variables in the estimates presented in Columns (4) and (5) test this hypothesis, and we identify increases in tubal ligation rates three to six months (one to two quarters) after the treatment date. These estimates reveal that around six months following a 100-mile increase in travel distance to an abortion provider, the rate of tubal ligation increases by 0.56 procedures per 10,000 females, a 9.1% increase from the sample mean.

Table 1: Effects of Abortion Travel Distance on Tubal Ligation Rate

	(1)	(2)	(3)	(4)	(5)
	tubal rate	tubal rate	tubal rate	tubal rate	tubal rate
Distance _{t+2}	0.114 (0.215)				
Distance ² _{t+2}	-0.034 (0.070)				
Distance _{t+1}		0.154 (0.224)			
Distance ² _{t+1}		-0.024 (0.074)			
Distance _t			0.286 (0.237)		
Distance ² _t			-0.048 (0.084)		
Distance _{t-1}				0.467* (0.282)	
Distance ² _{t-1}				-0.110 (0.098)	
Distance _{t-2}					0.555** (0.262)
Distance ² _{t-2}					-0.138 (0.093)
N	4820	4534	4795	4534	4280
y _{mean}	6.129	6.092	6.090	6.095	6.106

The observed 6-month lag represents a biological and logistical timeline where individuals become pregnant in the months leading up to HB2 and discover their pregnancies shortly after HB2 is passed. The immediate clinic closures following the passage of HB2 likely affected a cohort of individuals already in the first trimester of pregnancy, substantially increasing travel distance and reducing access to abortion services. Consequently, these individuals carry their pregnancy to term and choose sterilization immediately after giving birth to permanently insure against future pregnancy risk.

5.1 Event Study

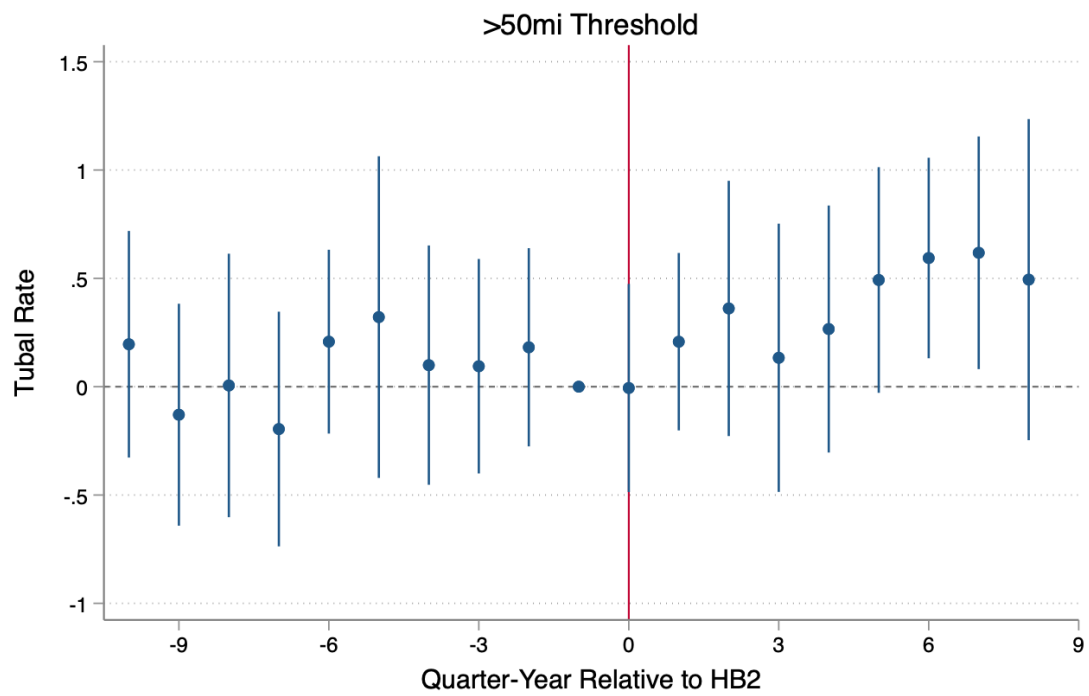
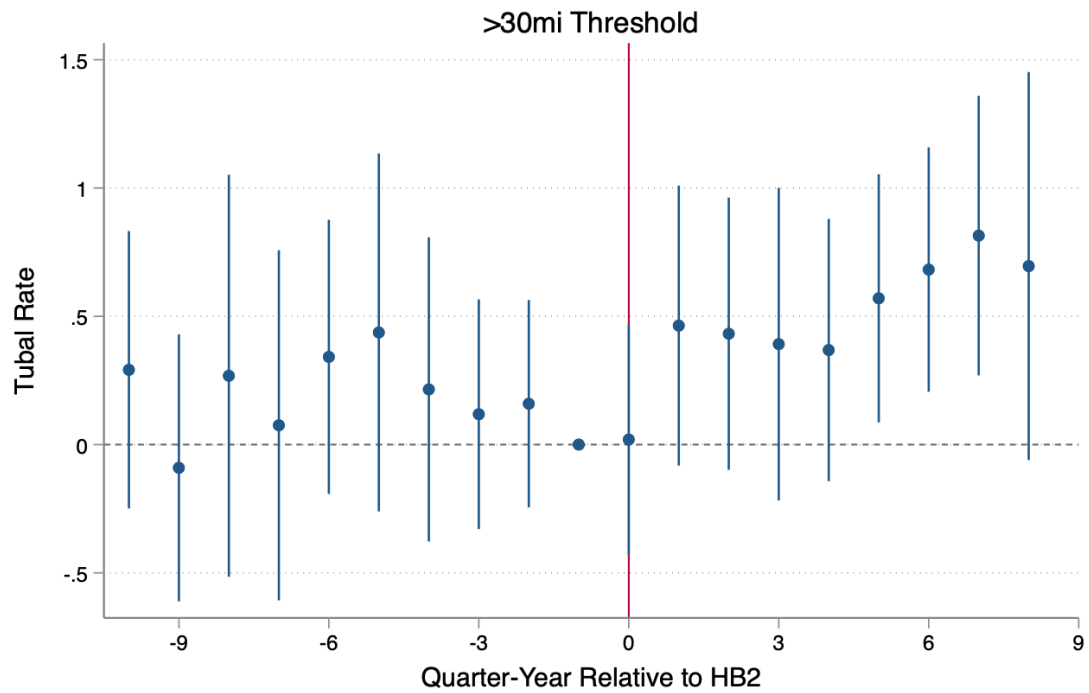
Table 3 demonstrates the importance of the dynamic effects of abortion access on tubal ligation rates, and therefore we complement the continuous difference-in-differences analysis with an event study design. In this analysis, we binarize the continuous measure of travel distance and estimate a standard linear event-study specification according to Equation (2),

$$Y_{ct} = \alpha_c + \delta_t + \sum_{k=-10}^8 \lambda_k Z_{c,t-k} + \beta X_{ct} + \epsilon_{ct} \quad (2)$$

where k indexes the coefficients λ relative to the passage of HB2 in the third quarter of 2013, and Z represents the binary indicator for county-level treatment by the policy change. We consider two possible thresholds for determining county treatment based on how large the travel distance change was post-HB2: a greater than 30-mile increase in travel distance and a greater than 50-mile increase in travel distance. For both thresholds, the comparison group is comprised of counties that experience less than a 30-mile increase in travel distance following the law. Therefore, counties with a 30-49-mile increase in travel distance are excluded from the analysis when using the 50-mile threshold for treatment.

Figure 4 plots the event study graphs for both treatment thresholds. Pre-trend coefficients suggest that tubal ligation trends were similar in the treatment and control counties, supporting the validity of the parallel trends' assumption. The trends in the post-HB2 period are consistent with our continuous difference-in-differences results in Table 3, pointing toward a lagged increase in the rate of tubal ligations after HB2. Once again, this suggests that tubal ligations increase in response to HB2 among the population of women realized a contraceptive failure, are unable to access an abortion, and subsequently undergo a postpartum sterilization at delivery.

Figure 4: Event Study – HB2 and Tubal Ligation Rate



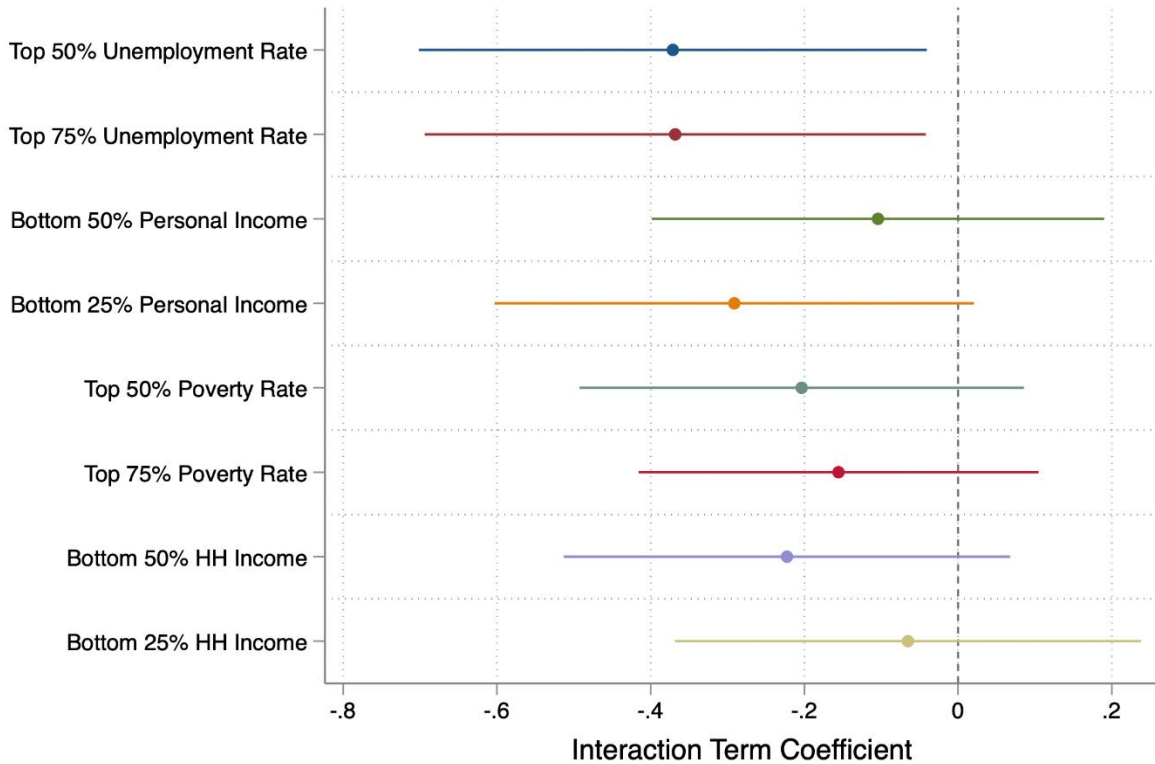
5.2 Heterogeneity Analysis by County Features

Restricted abortion access may differentially affect rates of tubal ligation among economically disadvantaged populations, but the direction of the effects is ambiguous ex-ante. On one hand, individuals experiencing economic disadvantage may place greater value on protection against contraceptive failure because they are less able to absorb the consequences of an unintended pregnancy. In this case, reduced abortion access could increase demand for tubal ligations among economically disadvantaged populations relative to other populations. Alternatively, economic insecurity may reduce access to healthcare and increase the barriers to obtaining procedures like tubal ligation. Under this mechanism, financially constrained individuals may be less able to obtain permanent sterilization despite increased demand.

To explore the nature of this intersection, we generate a series of county-level indicators reflecting the county's economic conditions. Specifically, we create binary indicators for counties in at the lower end of the distribution in terms of median and per capita personal income and at the higher end of the distribution in terms of poverty and unemployment rates. We repeat the estimation procedure in Equation (1), replacing the economic controls with the simple indicators and their interactions with our travel distance variable, lagged by two periods. The coefficient on the interaction term describes the effect of increasing travel distance by 100 miles on rates of tubal ligation in a county experiencing economic hardship.

Figure 5 provides a visual representation of the estimates. For most of the interactions, the effect is not statistically different from zero at the 5% level. However, the effects are all consistently negative, suggesting that living in economic hardship are less likely to acquire a tubal ligation following restricted access to abortion. This is consistent with similar evidence from Crowe et al. (2025), who find that Non-Hispanic White women and non-Medicaid recipients increase their take-up of IUDs the most after HB2.

Figure 5: Interaction of Economic Hardship and Travel Distance



5.3 Placebo Effects

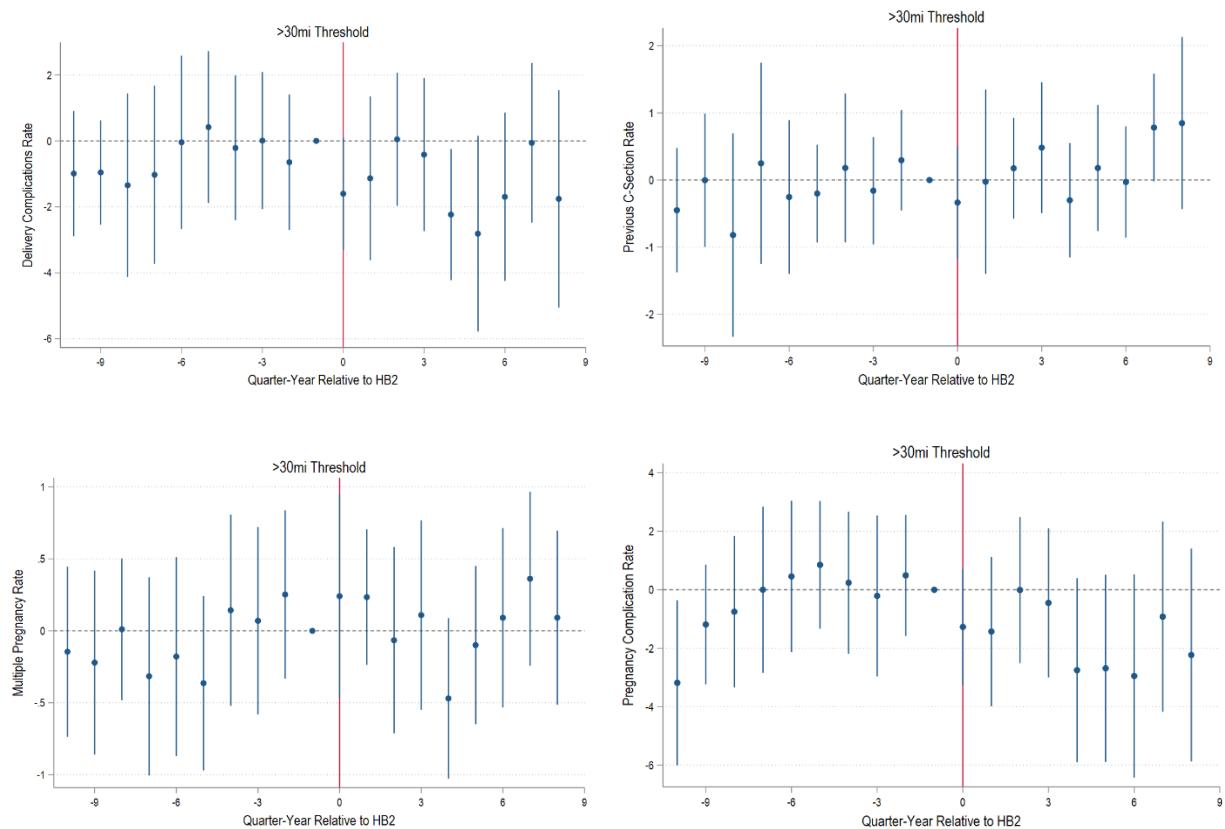
We also investigate whether increases in tubal ligation in affected counties could be explained by simultaneous increases in factors correlated with tubal ligation, such as patients having had a previous C-section, a pregnancy involving multiple fetuses, or complications during pregnancy or delivery. Albanese, French, and Gossett (2017) report that women with high-risk pregnancies are more likely to request a postpartum tubal ligation than those with low-risk pregnancies.

We used the ICD-9 diagnostic codes to create indicators of whether or not the patient experienced medical issues during pregnancy (including hypertension, pre-eclampsia, bleeding) or complications during delivery (including fetal complications, positioning issues, early birth). We also created indicators for whether or not the pregnancy involved multiple fetuses/births and whether or not the patient had had a previous C-section. As in the main analysis, we identify

these conditions if they appeared in any of the 26 diagnoses available for each patient, not just the principal diagnosis.

We repeat the event study procedure of the main analysis using as outcome variables the rates of health issues during pregnancy, complications during delivery, multiple pregnancies, and patients who had previously undergone a C-section. Results for the 30-mile treatment threshold are provided in Figure 6. Given positive correlation between sterilization and medical issues during pregnancy and delivery, we see these outcomes as potential mechanisms for increases in tubal ligation instance. However, we see no significant impact of increases in driving distance to abortion clinics on these potential mechanisms. This provides additional evidence that the increase in sterilizations post-HB2 that we found in the main analysis is motivated by desire to reduce unwanted pregnancies, not as a result of changes in related underlying medical conditions.

Figure 6: Placebo Analysis



6. Conclusion

With a wide range of available options and procedures, contraceptive decisions involve weighing risks, benefits, monetary costs, and the probability of failure. Female sterilization, including tubal ligation, serves as a highly effective contraceptive choice, but permanently removes the possibility of future children. This paper analyzes how demand for tubal ligation changes in response to reduced abortion access in Texas after the passage of HB2.

We first develop a theoretical framework to better understand how reductions in abortion access may lead to increases in female sterilization. If abortion serves as insurance against contraceptive failure, then increasing the cost of abortion access, or restricting it entirely, will lead individuals to choose more effective contraceptives to protect against pregnancy. This leads to two hypotheses about contraception choice. In one scenario, people with the capacity to become pregnant will choose tubal ligation proactively, leading to a sharp increase in procedures immediately after they lose access to abortion. However, tubal ligation alone is costly, involving surgical risks and time off work. In the second case, reduced access to abortion may also lead more women to choose tubal ligation after giving birth, when the costs of the procedure are lowest. Under this prediction, we expect to see a lagged increase in procedures as pregnancies are carried to full term.

We find that restricted abortion access following HB2 significantly increased rates of tubal ligation in the counties impacted by the legislation. Specifically, six months after travel distance to an abortion provider increased by 100 miles, tubal ligation rates increased by 9.1%. We follow this analysis with an event study, revealing that the largest effects of HB2 on tubal ligation rates occur more than one year after the policy is passed. Together, these results support the hypothesis that more women choose sterilization after they give birth, rather than choosing sterilization proactively.

Finally, we look at how the results vary alongside county-level economic conditions. We find that women in counties experiencing more economic hardship, including lower income, higher poverty rates, and higher unemployment, are less likely to acquire a tubal ligation after a change in travel distance. This finding aligns well with existing case studies demonstrating higher rates of sterilization for rural women with restricted resources (Clark and Levy, 2025) and no/public insurance (Borrero et al., 2006).

These findings suggest that after abortion restrictions are implemented, women primarily choose tubal ligation after they give birth. Our work aligns with prior findings that show increases in tubal ligations after *Dobbs* (Ellison et al., 2024), and further suggests that these increases are most likely concentrated among postpartum women. Given evidence of unmet demand for postpartum tubal ligations in Texas (Potter et al., 2021a), our work likely underestimates demand for these procedures, particularly among publicly insured women.

Ultimately, this work carries important implications for healthcare policy. This paper highlights that increases in sterilization resulting from abortion bans are delayed and frequently occur alongside delivery, rather than as a form of defensive contraception. When the safety net of abortion is removed, risk-averse agents opt for permanent, irreversible procedures. Our heterogeneity analysis aligns with Potter et al. (2021a), suggesting that administrative hurdles may prevent lower-income populations from adjusting to these policy shocks. Potter et al. (2021a) suggest opportunities to improve access to postpartum sterilization, such as physician discussions about contraceptive preferences throughout pregnancy and medical education on tubal ligation eligibility. Implementing these policy changes would allow more women to access tubal ligation in the lowest-cost setting (postpartum) and decrease unwanted pregnancies. Similarly, increasing education about the effectiveness of long-acting, reversible contraceptives (LARC) such as contraceptive implants and IUDs, would allow individuals to manage reproductive risk more effectively within a highly restricted policy environment.

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